Solution:

The p.m.f. Binomial distribution is

$p(x) = {}^nC_x p^x \quad q^{n-x}$

Given that $E(x) = 2$

For the Binomial distribution mean is given by $np = 2$ ... (1)

Given that var $(x) = 4/3$

For Binomial distribution variance is given by $npq = 4/3$ ... (2)

$$\frac{(2)}{(1)} \Rightarrow \frac{npq}{np} = \frac{\left(\frac{4}{3}\right)}{2} = \frac{4}{6} = \frac{2}{3}$$

$q = 2/3$ and $p = 1 - 2/3 = 1/3$

Substitute in (1) we get

$$n = 6$$

Hence, $P(X=5) = {}^6C_5 \left(\frac{1}{3}\right)^5 \left(\frac{2}{3}\right)^{6-5} = 0.0108$

Example 7.8

If the average rain falls on 9 days in every thirty days, find the probability that rain will fall on atleast two days of a given week.

Solution :

Probability of raining on a particular day is given by $p = 9/30 = 3/10$ and

$q = 1 - p = 7/10.$

The *binomial distribution* is

$P(X = x) = {}^nC_x p^x q^{n-x}$

There are 7 days in a week,

$$P(X = x) = \binom{7}{x}\left(\frac{3}{10}\right)^x \left(\frac{7}{10}\right)^{7-x}$$

The probability of raining for atleast 2 days is given by

$$P(X \geq 2) = 1 - P(X < 2)$$

$$= 1 - \left[P(X = 0) + P(X = 1)\right]$$

Here, $$P(X = 0) = \binom{7}{0}\left(\frac{3}{10}\right)^0 \left(\frac{7}{10}\right)^{7-0}$$

$$= 0.0823$$

and $$P(X = 1) = \binom{7}{1}\left(\frac{3}{10}\right)\left(\frac{7}{10}\right)^{7-1}$$

$$= 0.2471$$

Therefore the required probability

$= 1 - [P(x = 0) + P(x = 1)]$

$= 1 - \{0.082 + 0.247]$

$= 0.6706$

Example 7.9

What is the probability of guessing correctly atleast six of the ten answers in a TRUE/FALSE objective test?

Solution :

Probability p of guessing an answer correctly is $p = \frac{1}{2}$

$\Rightarrow \quad q = \frac{1}{2}$

Probability of guessing correctly x answers in 10 questions

$$P(X = x) = p(x) = {}^nC_x \, p^x q^{n-x}$$

$$= 10Cx\left(\frac{1}{2}\right)^x \left(\frac{1}{2}\right)^{10-x}$$

The required probability $P(X \geq 6)$

$= P(6) + P(7) + P(8) + P(9) + P(10)$

$$= \left(\frac{1}{2}\right)^{10}\left[10C_6 + 10C_7 + 10C_8 + 10C_9 + 10C_{10}\right]$$

$$= \left[\frac{1}{1024}\right][210 + 120 + 45 + 10 + 1]$$

$$= \frac{193}{512}$$

XII Std - Business Maths & Stat EM Chapter 7.indd 153 2/29/2020 11:34:21 AM